End-term Report: River Network Detection Project

Srinivasan M (IMT2021058) Siddharth Kothari (IMT2021019)
Sankalp Kothari (IMT2021028)

December 13, 2025

1 Summary of Accomplishments

During the second half of the semester, we successfully trained the model across various levels of corruption and generated detailed visualizations to support our findings in the paper. We have also completed the initial drafts of the manuscript and are currently focused on refining it through minor changes and improvements. The following sections provide a comprehensive overview of the work accomplished thus far.

1.1 Data Visualizations

The visualizations in Figure 3 provide a detailed analysis of the corruption process applied through morphological operations, specifically erosion and dilation, across varying corruption levels.

1.1.1 Erosion

The left boxplot illustrates the percentage of corrupted pixels caused by erosion for each corruption level. As observed:

- The percentage of corrupted pixels increases initially with the corruption level, showing proper trends up to approximately 20%.
- Beyond 20%, the number of corrupted pixels plateaus, as evident from the stagnant median and lower quartile values. This stagnation is due to the nature of erosion, which eliminates white pixels. When the white pixel count is low, further erosion becomes ineffective.
- This behavior demonstrates that erosion-based corruption is inherently limited by the white-pixel ratio in the original mask.

1.1.2 Dilation

The right boxplot showcases the percentage of corrupted pixels caused by dilation for each corruption level. Key observations include:

- The dilation process shows a more consistent trend, with a steady increase in the percentage of corrupted pixels as the corruption level increases.
- Unlike erosion, dilation's ability to corrupt does not stagnate, as it operates by expanding the white pixel regions, which are typically more abundant.
- This trend highlights the potential for dilation to introduce more uniform corruption compared to erosion.

These visualizations underline the differences between erosion and dilation in their corruption mechanisms. Erosion tends to be limited by the characteristics of the image, while dilation provides a more scalable corruption process across all levels.

1.2 SSIM

Figure 4 illustrates the distribution of Structural Similarity Index Measure (SSIM) scores across different corruption thresholds. SSIM is a perceptual metric that quantifies the similarity between two images, with higher values indicating greater structural similarity. The x-axis represents the corruption thresholds, expressed as percentages, while the y-axis corresponds to the SSIM scores.

At lower corruption thresholds (2% to 15%), the SSIM scores are clustered near 1.0, indicating minimal perceptual differences between the corrupted and original images. However, at the 2% threshold, there is noticeable variance with several outliers, reflecting higher sensitivity to minor corruptions. As the thresholds increase from 17% to 20%, the median SSIM scores begin to decline slightly, and the interquartile range expands, indicating a broader distribution of perceptual differences caused by corruption.

At higher thresholds (25% and 30%), the SSIM scores show a more significant spread, with a broader interquartile range and lower whiskers, suggesting that corruption has a more pronounced effect on some images. This trend demonstrates that higher corruption levels cause varying degrees of perceptual degradation across the dataset. The overall trend shows a gradual decline in structural similarity as the severity of corruption increases, highlighting the impact of morphological operations on image quality.

1.3 Observations and Results

This section provides an analysis of the model's performance across various corruption levels, including quantitative metrics and qualitative insights, to evaluate the robustness of the U-Net architecture.

1.3.1 Quantitative Performance Analysis

The evaluation metrics for the U-Net model across varying corruption levels are summarized in Table 1. Under standard conditions, the model achieved a high segmentation accuracy of 95%. Remarkably, this performance remained robust, with minimal degradation, even as the corruption levels increased to 20%. Such stability highlights the model's ability to tolerate high levels of corruption effectively.

The results reveal that the model can maintain strong performance up to 25% corruption before significant degradation occurs. To investigate further, we examined the model's training process, qualitative predictions, and the nature of the corruptions it encountered.

Figure 1 depicts the accuracy trends per epoch for training and validation datasets at different corruption levels. Validation accuracy remained stable up to 17% corruption but began showing instability beyond this threshold. Importantly, the model resisted overfitting, as seen in the declining training accuracy with increasing corruption levels, while validation accuracy remained consistent.

1.3.2 Qualitative Insights and Robustness

Figure 2 provides visual examples of the U-Net predictions. The input images, corrupted masks (with erosion applied), and predicted outputs illustrate the model's robustness under varying corruption levels.

For images with significant erosion, the percentage of white pixels in the masks decreased substantially, with no remaining white regions at 25% corruption. Interestingly, increasing corruption did not always proportionally increase the number of corrupted pixels, particularly for images with sparse foreground features.

Despite this, the U-Net output before thresholding appeared blurred with higher corruption levels, indicating difficulty in distinguishing land from water. However, the thresholded outputs maintained reasonable accuracy in identifying water-body structures, showcasing the model's ability to handle significant corruption effectively.

1.3.3 Performance Trends at High Corruption Levels

The results indicate a clear corruption threshold. The model maintained stable validation performance up to 17% corruption, beyond which performance began to decline. Notably, even at 20% corruption, the thresholded predictions remained reasonably accurate in identifying river structures. However, predictions became increasingly blurred and less reliable beyond 25% corruption.

These findings emphasize the U-Net architecture's inherent resilience to noise and imperfections in input data. The model effectively generalized under challenging conditions, avoiding overfitting and retaining segmentation accuracy. Nonetheless, extreme corruption levels (¿25%) highlighted the model's limitations in discerning fine-grained details.

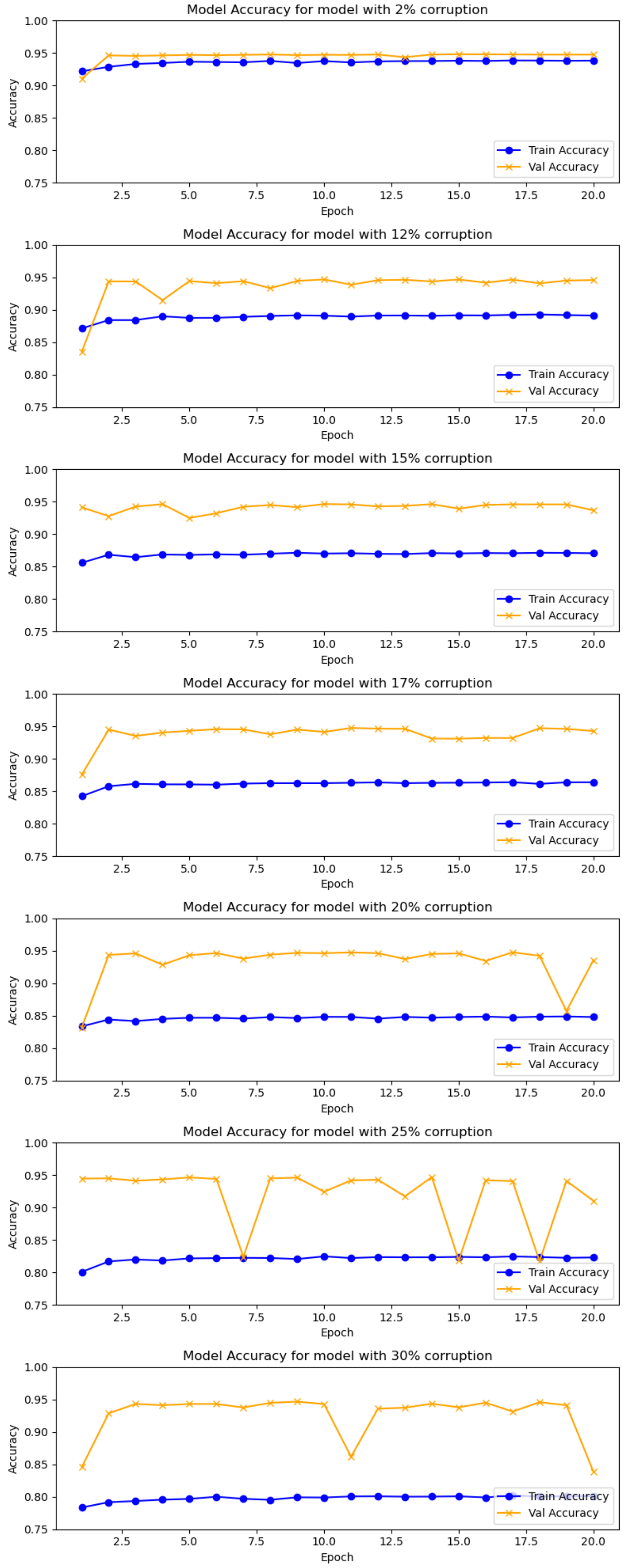

Figure 1: Training and Validation Accuracy Plots

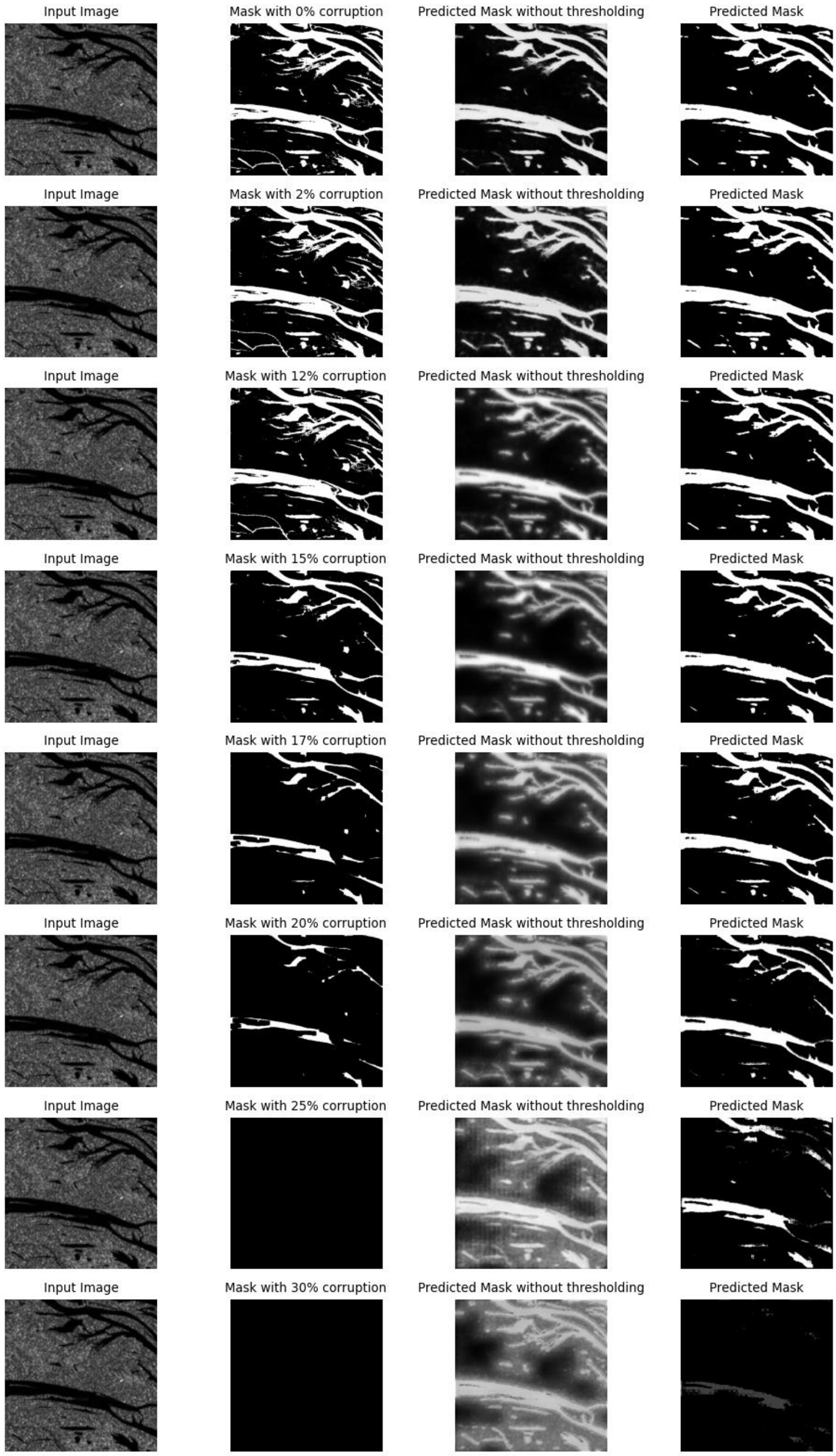

Figure 2: Qualitative Plot showing the effect of corruption level on the predicted and actual masks for one datapoint

Table 1: Performance Metrics of U-Net Model at Various Corruption Levels

Metric	Clean	2%	12%	15%	17%	20%	25%	30%
Average Dice Coefficient	0.7696	0.7748	0.7693	0.7678	0.7580	0.7252	0.5971	0.2532
Average IoU	0.8230	0.8252	0.8224	0.8190	0.8141	0.7897	0.7081	0.4964
Average Precision	0.8959	0.8959	0.8989	0.8899	0.9028	0.9024	0.8746	0.8178
Average Recall	0.8688	0.8734	0.8641	0.8663	0.8536	0.8303	0.7592	0.5878
Average F1 Score	0.8652	0.8677	0.8646	0.8633	0.8591	0.8405	0.7678	0.5672
Average Specificity	0.9708	0.9680	0.9748	0.9705	0.9785	0.9812	0.9842	0.9963
Model's Average Pixel Accuracy	0.9554	0.9546	0.9546	0.9541	0.9523	0.9460	0.9196	0.8282

1.4 Training of adversarially robust models

To establish a performance benchmark, we began by training a baseline model using a vanilla U-Net architecture. The dataset consisted of Sentinel-1 satellite imagery, and the ground truth annotations were generated water masks. This setup provided a foundation to evaluate the model's capability in handling subsequent experimental conditions.

The dataset was divided into training, validation, and testing subsets with a 70:10:20 split. The model was trained for 20 epochs across all experiments, ensuring consistency in the evaluation process.

To simulate human-introducible errors that may occur during manual annotation, we introduced controlled corruptions to the water masks using two fundamental morphological operations: erosion and dilation. These operations are commonly used in image processing to modify the structure of binary or grayscale images by analyzing pixel neighborhoods, defined by a kernel size.

Erosion: In this operation, each pixel in the output image takes the minimum value of all the pixels in its neighborhood. For binary images, a pixel is set to 0 if any neighboring pixel has a value of 0. This results in the shrinking of white regions (foreground) in the image, effectively simulating scenarios where features may appear diminished due to annotation errors.

Dilation: In dilation, each pixel in the output image takes the maximum value of all the pixels in its neighborhood. For binary images, a pixel is set to 1 if any neighboring pixel has a value of 1. This expands the white regions (foreground), mimicking scenarios where features may be exaggerated during manual annotation.

The corruption methodology involved dividing the training and validation sets into three equal parts: one-third of the images were corrupted using erosion, another third using dilation, and the remaining third were left uncorrupted. Corruptions were applied iteratively to masks until either a maximum of 100 iterations was reached or the number of corrupted pixels exceeded a predefined threshold. If the threshold was surpassed, the mask from the previous iteration was retained.

To introduce variability and mitigate potential biases in the model's learning, the kernel sizes used for erosion and dilation were chosen stochastically. The kernel size depended on the ratio of white pixels (foreground) in the image. Specifically, images with fewer white pixels used larger kernels for erosion, while smaller kernels were used for dilation. This stochasticity ensured the corruptions were non-uniform, adding diversity to the corrupted dataset.

2 Conclusion

- In this work, we proposed a novel approach to simulate human-introduced errors in segmentation masks using morphological operations, specifically erosion and dilation, to introduce controlled corruptions.
- The U-Net model demonstrated robustness against these morphological corruptions, maintaining high accuracy in detecting water structures even under significant corruption levels, up to a defined corruption threshold.
- The findings highlight the model's ability to generalize and remain effective under noisy and imperfect conditions, showcasing its resilience in real-world scenarios where manual annotation errors may occur.
- Future directions of research include extending the study to incorporate other forms of corruption, such as geometric transformations (e.g., rotation and translation), to evaluate their impact on model performance and robustness.

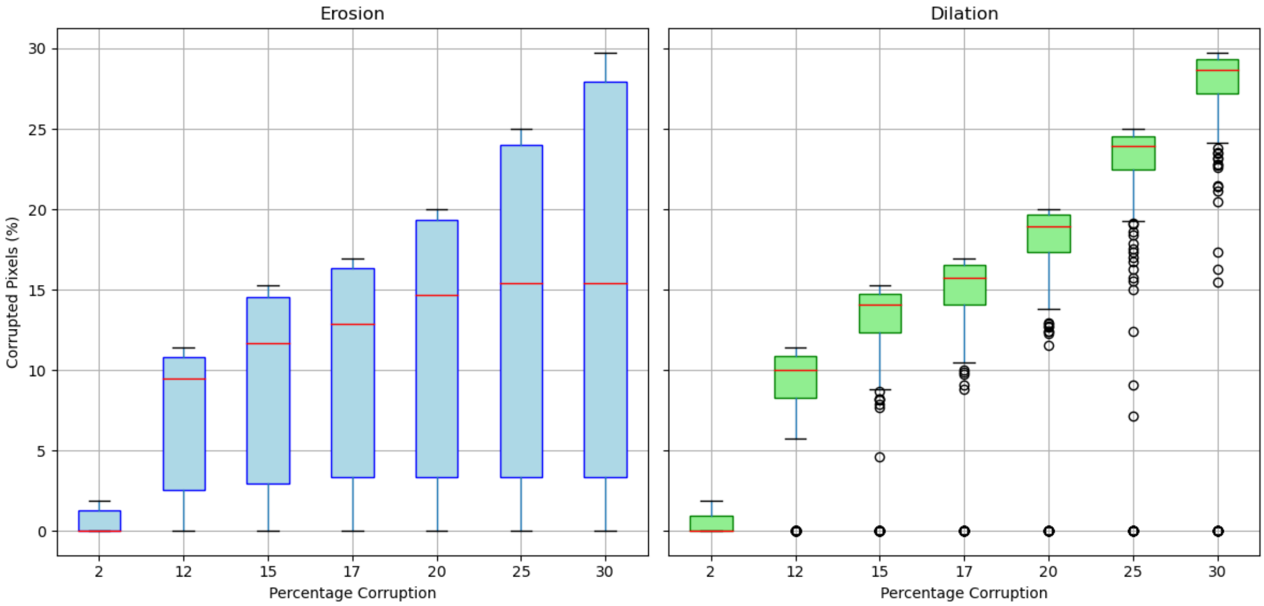

Figure 3: Boxplots showing the actual corruption versus the threshold

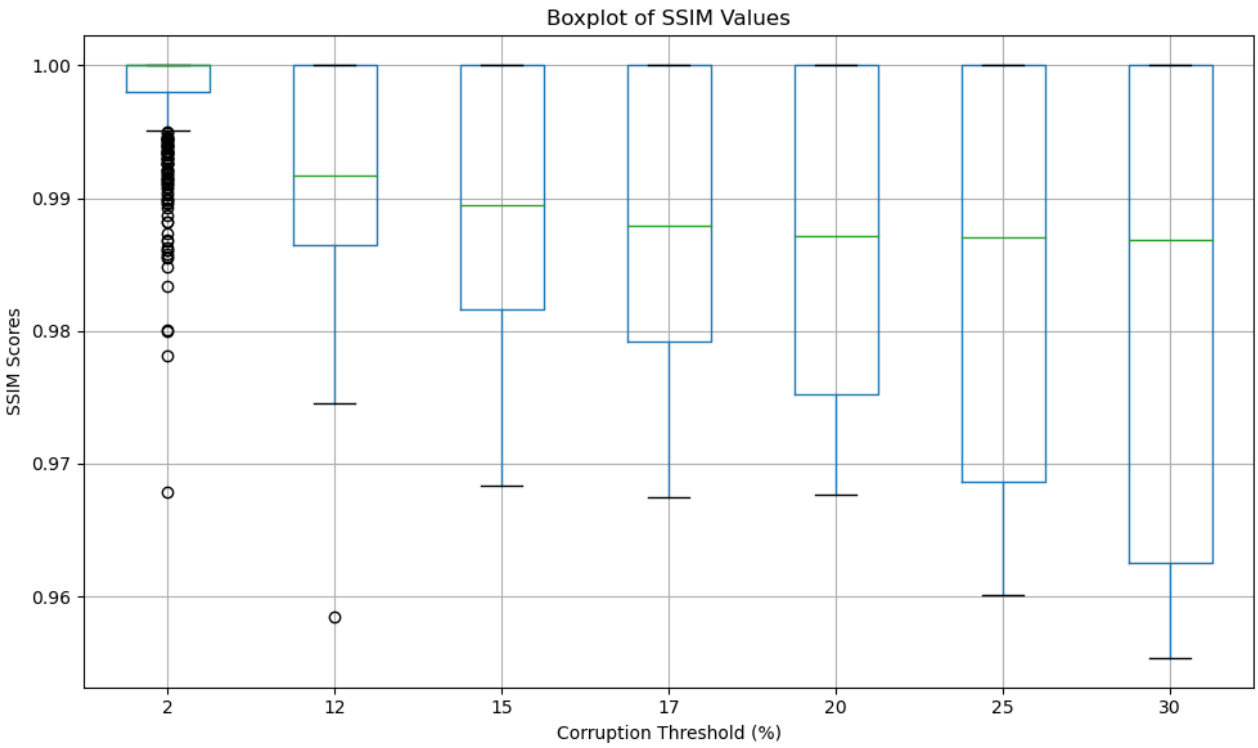

Figure 4: Boxplots showing the SSIM values between the actual mask and the corrupted masks for various levels of corruption

References

[1] Obida, C. B., Blackburn, G. A., Whyatt, J. D., and Semple, K. T., "River network delineation from Sentinel-1 SAR data", *International Journal of Applied Earth Observation and Geoinformation*, vol. 83, 2019. doi:10.1016/j.jag.2019.101910.

[2] Code to bulk download data from Alaska Satellite Facility - https://www.linkedin.com/pulse/bulk-download-sentinel-1-data-aditya-sharma/

[3] Performing preprocessing on SAR images using Python library Snappy - https://step.esa.int/docs/tutorials/Performing%20SAR%20processing%20in%20Python%20using%20snappy.pdf

[4] Snappy installation guide - https://gitlab.com/JohnCrabs/installation_esa_snappy

[5] M. Wieland et al., "S1S2-Water: A Global Dataset for Semantic Segmentation of Water Bodies From Sentinel- 1 and Sentinel-2 Satellite Images," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, pp. 1084-1099, 2024, doi: 10.1109/JSTARS.2023.3333969.

[6] D. Bonafilia, B. Tellman, T. Anderson and E. Issenberg, "Sen1Floods11: a georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1," 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Seattle, WA, USA, 2020, pp. 835-845, doi: 10.1109/CVPRW50498.2020.00113.

[7] ESA SNAP Documentation - https://step.esa.int/main/doc/

[8] Numpy Documentation - https://numpy.org/doc/

[9] Scikit Learn Documentation - https://scikit-learn.org/

[10] Matplotlib Documentation - https://matplotlib.org/stable/index.html

[11] OsGeo Documentation - https://gdal.org/api/python/osgeo.html

[12] Geopandas Documentation - https://geopandas.org/en/stable/docs.html

[13] ASF_Search Documentation - https://docs.asf.alaska.edu/asf_search/basics/

[14] Copernicus Browser Data Download Website - https://dataspace.copernicus.eu/browser/?zoom=8&lat=-1.93853&lng=-54.25598&themeId=DEFAULT-THEME&visualizationUrl=https%3A%2F%2Fsh.dataspace.copernicus.eu%2Fogc%2Fwms%2Fea820656-1f43-408b-9554-afc79eea489a&datasetId=S1_CDAS_IW_VVVH&fromTime=2023-11-14T00%3A00%3A00.000Z&toTime=2023-11-14T23%3A59%3A59.999Z&layerId=8_RGB-RATIO

[15] Shapely Documentation - https://shapely.readthedocs.io/en/stable/manual.html

[16] KMeans Clustering implementation on Sentinel 1 and 2 - https://github.com/acgeospatial/Satellite_Imagery_Python/blob/master/Clustering_KMeans-Sentinel2.ipynb

[17] Y. Qi, Y. He, X. Qi, Y. Zhang and G. Yang, "Dynamic Snake Convolution based on Topological Geometric Constraints for Tubular Structure Segmentation," 2023 IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2023, pp. 6047-6056, doi: 10.1109/ICCV51070.2023.00558.

[18] Ujjwal Verma, Arjun Chauhan, Manohara Pai M.M., Radhika Pai, DeepRivWidth : Deep learning based semantic segmentation approach for river identification and width measurement in SAR images of Coastal Karnataka, Computers & Geosciences, Volume 154, 2021, 104805, ISSN 0098-3004, https://doi.org/10.1016/j.cageo.2021.104805.

[19] Guo, Z.; Wu, L.; Huang, Y.; Guo, Z.; Zhao, J.; Li, N. Water-Body Segmentation for SAR Images: Past, Current, and Future. Remote Sens. 2022, 14, 1752. https://doi.org/10.3390/rs14071752

[20] Wang, D.; Cao, W.; Zhang, F.; Li, Z.; Xu, S.; Wu, X. A Review of Deep Learning in Multiscale Agricultural Sensing. Remote Sens. 2022, 14, 559. https://doi.org/10.3390/rs14030559 h

[21] HydroSHEDS HydroRivers Dataset, available at https://www.hydrosheds.org/products/hydrorivers

[22] LBA-ECO CD-06 Amazon River Basin Land and Stream Drainage Direction Maps, available at https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=1086